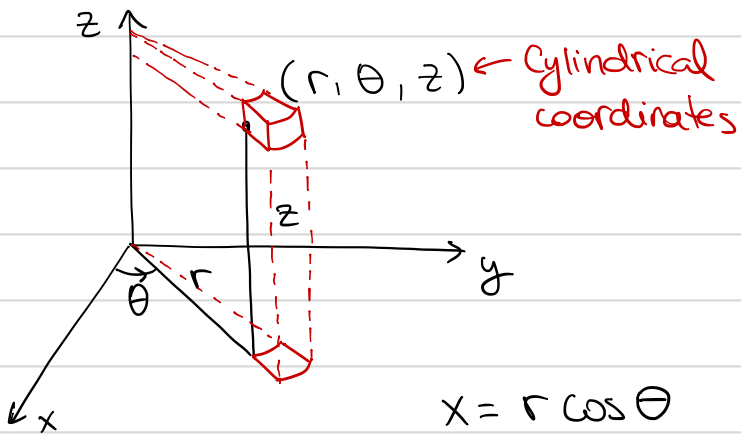
Cylindrical coordinates (§15.7)



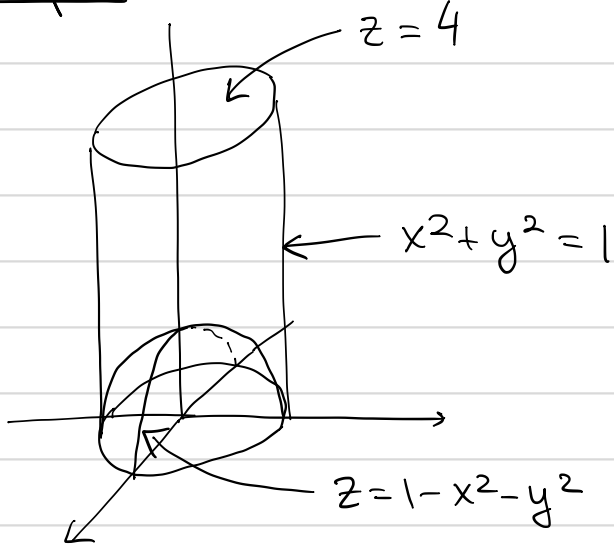
$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

Volume element: $dV = r\,dr\,d\theta\,dz$

Example:



Density: $\rho = Kz$

$$\text{Mass} = \iiint_E \rho\, dV$$

$$= \int_0^{2\pi}\int_0^1\int_{z=1-r^2}^{4} Kz\, dz\, r\,dr\, d\theta$$

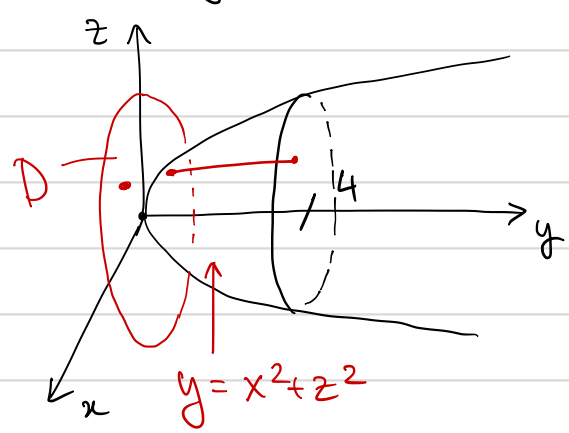
$$= 2\pi K \int_0^1 \underbrace{\tfrac{1}{2}\left(16-(1-2r^2+r^4)\right)}_{\frac{15}{2}+r^2-\frac{1}{2}r^4} r\,dr$$

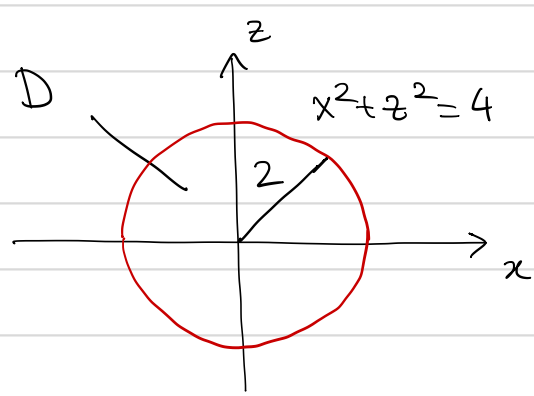
$$= 2\pi K \left[\frac{15}{4}r^2 + \frac{r^3}{3} - \frac{1}{10}r^5\right]_{r=0}^{1}$$

$$= 2\pi K\left(\frac{15}{4}+\frac{1}{3}-\frac{1}{10}\right) \;//$$

← skip in class

Example: E = region between $y = x^2+z^2$ and $y=4$





$$\iiint_E \sqrt{x^2+z^2}\, dV =$$

$x = r\cos\theta$

$z = r\sin\theta$

$y = y$

$$= \iint_D \int_{y=r^2}^{4} r \, dy \, dA$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{2} (4-r^2)\, r \cdot r \, dr \, d\theta$$

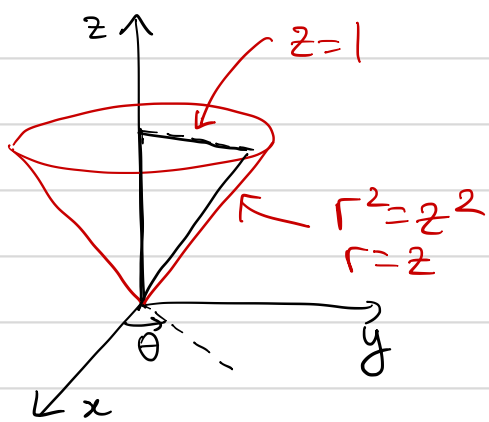
$$= 2\pi \left[\frac{4r^3}{3} - \frac{r^5}{5} \right]_0^2 = 2\pi \left[\frac{32}{3} - \frac{32}{5} \right]$$

$$= \frac{128}{15}\pi \;//$$

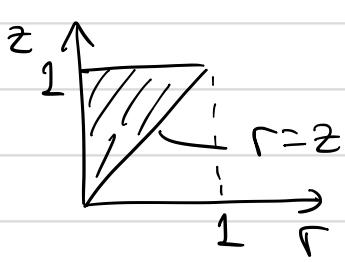
<u>Example</u>: Moment of inertia about z-axis of solid cone bounded by

$x^2+y^2=z^2$ and $z=1$.

Uniform density: $\rho = 1$.



θ-slice:



$$I_z = \iiint_E (x^2+y^2)\, \rho \, dV$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{1} \int_{z=r}^{1} r^2 \cdot 1 \cdot r \, dz \, dr \, d\theta$$

$$= 2\pi \int_0^1 r^3 (1-r)\, dr$$

$$= 2\pi \left(\frac{r^4}{4} - \frac{r^5}{5} \right) \Big|_0^1$$

$$= 2\pi \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{\pi}{10} \;//$$

<u>Example</u>: Volume between:

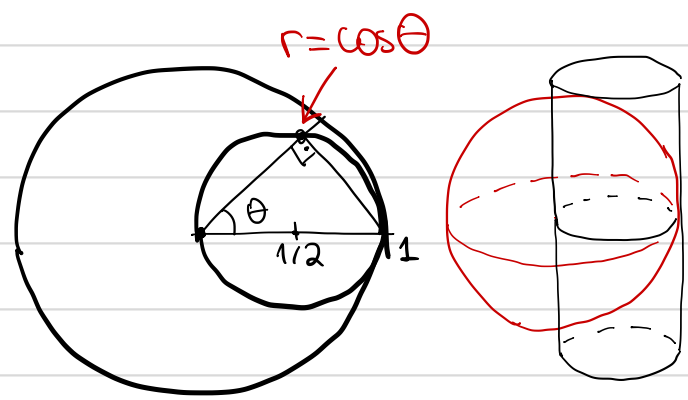
$x^2+y^2+z^2=1$ (sphere)

$r = \cos\theta$ (cylinder)

$$\left.\begin{aligned} r^2 &= r\cos\theta \\ x^2+y^2 &= x \\ (x-\tfrac{1}{2})^2 + y^2 &= \tfrac{1}{4} \end{aligned}\right\}$$

Center: $(\frac{1}{2}, 0)$

Radius: $\frac{1}{2}$



$$\int_{\theta=-\frac{\pi}{2}}^{\pi/2}\int_{r=0}^{\cos\theta}\int_{z=-\sqrt{1-r^2}}^{\sqrt{1-r^2}} dz\ r\ dr\ d\theta$$

$$=\int_{-\pi/2}^{\pi/2}\int_{0}^{\cos\theta} 2\sqrt{1-r^2}\,r\,dr\,d\theta$$

$$=\int_{-\pi/2}^{\pi/2}\left[-\frac{(1-r^2)^{3/2}}{3/2}\right]_{r=0}^{\cos\theta} d\theta$$

$$=-\frac{2}{3}\int_{-\pi/2}^{\pi/2}(\underbrace{1-\cos^2\theta}_{\sin^2\theta})^{3/2}-1\ d\theta$$

$$=-\frac{2}{3}\int_{-\pi/2}^{\pi/2}\underbrace{|\sin\theta|^3-1}_{\text{even function}}\ d\theta$$

$$=-\frac{4}{3}\int_{0}^{\pi/2}\underbrace{\sin^3\theta}_{(1-\cos^2\theta)\sin\theta}-1\ d\theta$$

$$=-\frac{4}{3}\left[-\cos\theta+\frac{\cos^3\theta}{3}-\theta\right]_0^{\pi/2}$$

$$=-\frac{4}{3}\left[\underbrace{-0+0-\frac{\pi}{2}-\left(-1+\frac{1}{3}-0\right)}_{-\pi/2+2/3}\right]$$

$$=\boxed{\frac{2}{3}\pi-\frac{8}{9}}$$